

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA
IIT - JEE ACADEMY - INDIA

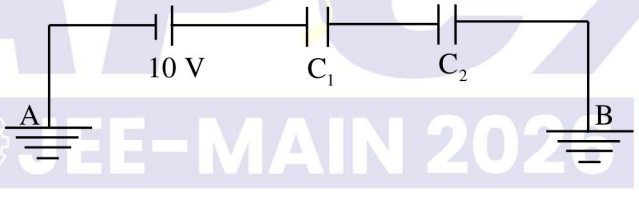
Sec: JEE APEX & SR IIT MODEL-A **DT: 31/12/2025**
Time: 3 Hrs **GTM - 06** **Max. Marks: 300**

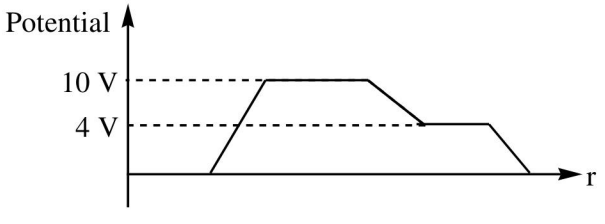
PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. Figure shows two capacitors C_1 and C_2 connected with 10V battery and terminal A and B are earthed. The graph shows the variation of potential as one moves from left to right. Then the ratio C_1 / C_2 is





A) 5/2 B) 2/3 C) 2/5 D) 4/3

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2. The velocity of a small ball of mass m and density d_1, when dropped in a container filled with glycerin (density d_2) is constant then consider the following statement regarding viscous force and velocity of the ball.

a) viscous force will depend upon the ratio of d_1/d_2

b) viscous force is dependent upon the mass of ball

c) Terminal velocity will not depend upon radius of ball

d) viscous force doesn`t depend upon the mass of ball

e) Terminal velocity will depend upon d_1 & d_2

Choose the correct as:

A) a, b and e are correct

B) c, d and e are correct

C) a, d and e are correct

D) b, c and e are correct

3. Consider the following statements

a) Liquids with strong cohesive forces will have lesser surface tension

b) Viscosity generally increases with pressure because molecules are forced closer

c) Coefficient of Viscosity also depends upon area of contact between layers and distance between layers

d) Reynold`s number for any type of flow increases with increment in density of liquid.

e) For laminar flow ;value of Reynold`s number will be less in comparison with turbulent flow.

Choose the correct answer from option given below

A) b, c, d and e are correct

B) b, d and e are correct

C) only a and c are correct

D) only b is correct

4. For a given region

$\vec{E} = yz\hat{i} + xz\hat{j} + xy\hat{k}$

Then the equation of equipotential surface will be: [Assume potential is zero at origin]

A) $xyz = \text{constant}$

B) $x + y + z = \text{constant}$

C) $xy + yz + xz = \text{constant}$

D) $x^2 + y^2 + z^2 = \text{constant}$

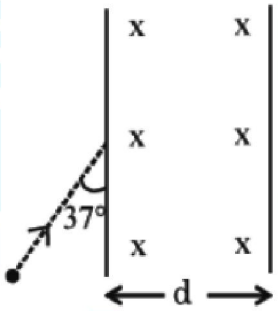
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5. A coin is placed at the bottom of an empty hemispherical bowl of radius R and is just not visible to an eye looking over the edge. When the bowl is filled with water the whole of the coin is just visible to an eye in the same position. The diameter of the coin; if μ is the refractive index of water is:



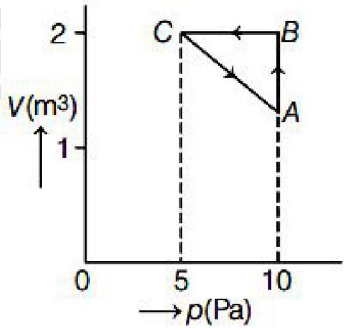
A) $\frac{2R\left(\mu^2-1\right)}{\mu^2+1}$ B) $\frac{R\left(\mu^2+1\right)}{\mu^2-1}$ C) $R\left(\frac{\mu^2-1}{\mu^2+1}\right)$ D) $\frac{R}{2}\left(\frac{\mu^2-1}{\mu^2+1}\right)$

6. A particle of mass 1 mg and charge 1 μc is projected with speed 50 m/s towards a region of magnetic field 10 Tesla containing a minimum thickness (d) of this region (in m) so that particle returns back to left side.



A) 1 m B) 1.2 m C) 0.7 m D) 1.50 m

7. An ideal gas is taken through the cycle $A \to B \to C \to A$, as shown in figure. If the net heat supplied to the gas in cycle is $5J$, work done by the gas in the process $C \to A$



A) $-5J$ B) $-10J$ C) $15J$ D) $-20J$

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8. Given below are two statements: one is labelled as Assertion and other is labelled as Reason

Assertion (A):-The speed of sound is greater on a hot summer day than on a cold winter day

Reason (R):-The velocity of sound in a gas is directly proportional to the square root of it`s absolute temperature.

Choose the correct answer from the option given below

A) Both A and R are true and R is the correct explanation of A.

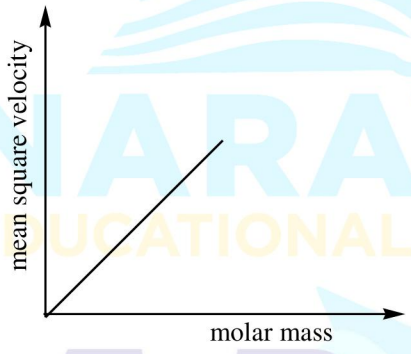
B) A is false but R is true

C) Both A and R are true but R is not the correct explanation of A

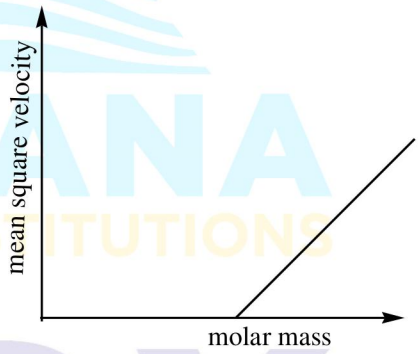
D) A is true but R is false.

9. The graph which represents the variation of mean square velocity of molecules with molar mass is

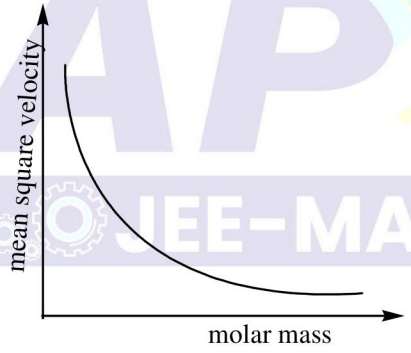
A)



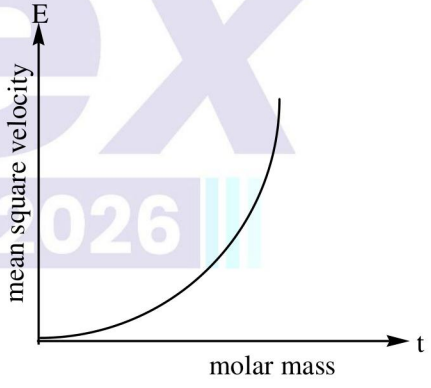
B)



C)

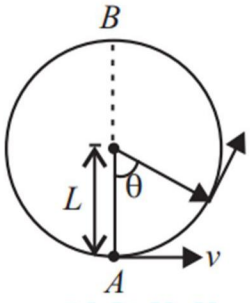


D)



10. A bob of mass M is suspended by a massless string of length L. the horizontal velocity V at position A is just sufficient to make it reach the point B. The angle θ at which the speed of the bob is half of that at A, satisfies

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A) $\theta = \frac{\pi}{4}$ B) $\frac{\pi}{4} < \theta < \frac{\pi}{2}$ C) $\frac{\pi}{4} < \theta < \frac{3\pi}{4}$ D) $\frac{3\pi}{4} < \theta < \pi$

11. Given below are two statements; one is labelled as Assertion and other is labelled as reason

Assertion (A):- Work done by friction over a round trip is non zero.

Reason (R):- Friction is a non-conservative force, and it`s work depends on the path taken.

Choose the correct answer from the option given below

A) Both A and R are true and R is the correct explanation of A

B) A is false but R is true

C) Both A and R are true but R is not the correct explanation of A

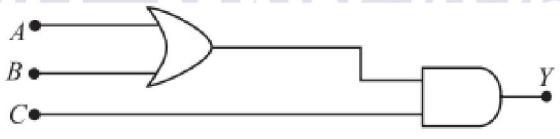
D) A is true but R is false

12. A radioactive nucleus ${}^{A}_{Z}X$ undergoes a series of decays, emitting four α particles and two β^- particles. The final stable nucleus formed is ${}^{206}_{82}Pb.$. What are the mass number (A) and atomic number (Z) of the original nucleus ${}^{A}_{Z}X$

A) A=222, Z=90 B) A=238, Z=92

C) A=232, Z=90 D) A=222, Z=88

13. To get an output Y=1 from circuit of figure the inputs must be



	A	B	C
A)	0	1	0
C)	1	0	0

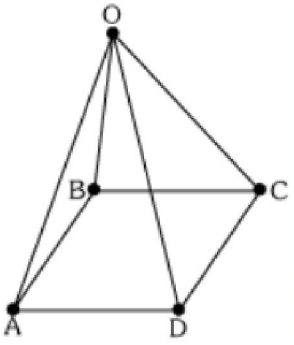
	A	B	C
B)	0	0	1
D)	1	0	1

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14. A piece of conducting wire of resistance R is cut into 2n equal parts. Half of the parts are connected in series to form a bundle and remaining half in parallel to form another bundle. These bundles are then connected to give the maximum resistance. The maximum resistance of the combination is

A) $\frac{R}{2}\left(1+\frac{1}{n^2}\right)$ B) $\frac{R}{2}\left(1+n^2\right)$ C) $\frac{R}{2\left(1+n^2\right)}$ D) $R\left(n+\frac{1}{n}\right)$

15. Eight identical resistance r each are connected along edges of a pyramid having square base ABCD as shown in figure.



Choose the correct statement

A) The equivalent resistance between A and D is $\frac{6r}{15}$

B) The equivalent resistance between A and D is $\frac{r}{5}$.

C) The equivalent resistance between A and D is $\frac{8r}{15}$

D) The equivalent resistance between A and D is $\frac{11r}{15}$

16. The electric field of the emerging beam from filter is represented by:

$E_x = 6\sin\left(1.20\times10^7 z - 3.6\times10^{13} t\right)$

Then find out equations of electric field in a region where energy density of E.M wave is two times in filter

A) $6\sqrt{2}\ \sin\left(1.20\times10^7 z - 3.6\times10^{15} t\right)$ B) $5\sin\left(1.20\times10^7 z - 3.6\times10^{13} t\right)$

C) $6\sin\left(1.20\times10^7 z - 3.6\times10^{15} t\right)$ D) $4\sin\left(1.20\times10^7 z - 3.6\times10^{19} t\right)$

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17. Two identical thin rings, each of radius a are placed coaxially at a distance "a" apart. Let charges Q_1 and Q_2 be placed uniformly on the two rings. The work done in moving a change Q from the centre of one ring to that of the other is

A) zero

B) $\frac{Q\sqrt{2}}{4\pi \in_0 a}(Q_1 + Q_2)$

C) $\frac{Q(\sqrt{2}-1)}{4\pi \in_0 a\sqrt{2}}(Q_1 - Q_2)$

D) $\frac{Q(\sqrt{2}+1)}{4\pi \in_0 a}(Q_1 - Q_2)$

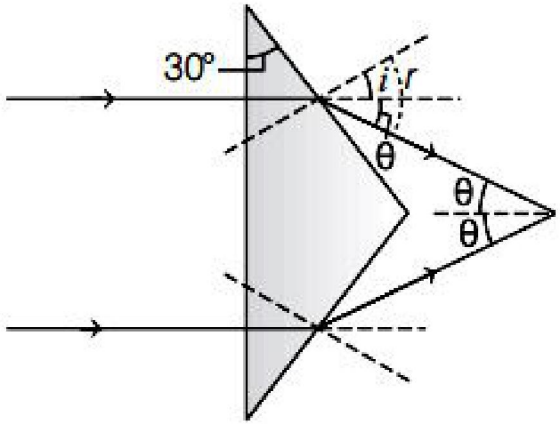
18. Find out the ratio of de-Broglie wavelength of a proton and an α particle which have been accelerated through same potential difference

A) $2\sqrt{2}:1$ B) $3:2$ C) $3\sqrt{2}:1$ D) $2:1$

19. A uniform thin rod AB of length L has linear mass density $\mu(x) = a + \frac{bx}{L}$ where μ is measured from A. If the C.O.M of the rod lies at a distance of $\frac{7L}{12}$ from A then a and b are related as

A) a = b B) a = 2b C) 2a = b D) 3a = 2b

20. Two parallel light rays are incident at one surface of a prism of refractive index 1.5 as shown in figure. The angle between the emergent rays is nearly



A) 19^0 B) 37^0 C) 45^0 D) 49^0

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SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

21. A spring is placed between the jaws of gauge such that the spring is not at all compressed the main scale reads 2 division and circular scale read 28 divisions. Now we turn the circular scale by 18^0 such that the spring is compressed. The circular scale has 200 divisions and the least count of the main scale is 1mm. What is the force exerted by the spring on the jaws if the spring constant is 100N/m. Give your answer in mN.

22. Let the moment of inertia of a hollow cylinder of length 30 cm (inner radius 10 cm and outer radius 20cm) about it`s axis be I. The radius of a thin hollow cylinder of the same mass such that it`s moment of inertia about it`s axis is also I; is
(Give nearest integer answer in cm)

23. If the dimensional formula for angular momentum per unit mass is $M^aL^bT^c$ then a + b + c will be

24. Given a uniform disc of mass M and radius R. A small disc of radius R/2 is cut from this disc in such a way that the distance between the centres of the two discs is R/2. If the moment of inertia of the remaining disc about a diameter of the original disc perpendicular to the line connecting the centres of the two discs is $x\frac{MR^2}{64}$, then the value of 'x' is equal to

25. In young`s double slit experiment, slits are separated by 2mm and the screen is placed at a distance of 1.2m from the slits. Lights consisting of wavelength 6500 $\overset{0}{A}$ and 5200 $\overset{0}{A}$ are used to obtain interference fringes. The separation between the fourth bright fringes of two different putters produced by the two wavelength is equal to (in μ m)

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SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Regarding Boron family the wrong statement is

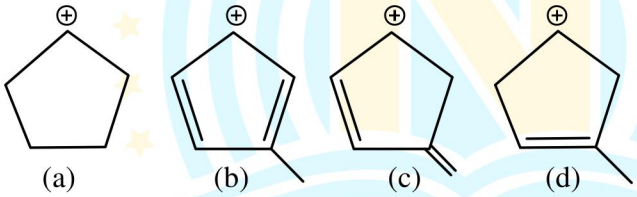
A) The difference in boiling point to melting point is highest for Gallium

B) Boron has highest melting point

C) Thallium has least boiling point

D) Ionization enthalpy of Indium is more than that of Thallium

27. Among the following select the least and most stable carbocations respectively

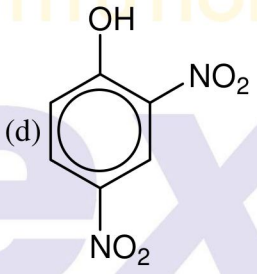


A) b and c B) c and d C) c and b D) a and b

28. The compounds that produce CO_2 with aqueous $NaHCO_3$ solution are

(a) $Ph-CH_2-OH$

(b) C_6H_5-COOH



(d)

(c) carbolic acid

(e) CH_3-CO_2H

Choose the correct answer from the options given below:

A) b and d only B) a, c, d and e only

C) b, d and e only D) b, c & e only

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29. Regarding $KMnO_4$, Choose the correct statements given below

(a) It is prepared by fusion of MnO_2 with an alkali metal hydroxide and an oxidizing agent like KNO_3

(b) π-bonding takes place by overlap of p-orbitals of oxygen with d-orbitals of manganese

(c) It is isostructural with $KClO_4$

(d) In redox titrations involving acidified $KMnO_4$, HCl is not used since it is oxidized to chlorine

A) a, b and d only B) b, c, d only

C) a, b only D) a, b, c and d only

30. Statement – I: Aldohexose on prolonged heating with hydrogen iodide gives n-heptane

Statement – II: Glycosidic linkage between C_1 of α - (D) - glucose and C_2 of β - (D) - fructose is found in sucrose

A) Both statement-I and statement-II are false

B) Statement-I is true but statement-II is false

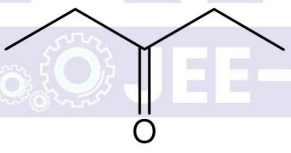
C) Both statement-I and statement-II are true

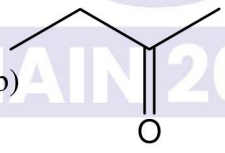
D) Statement-I is false but statement-II is true

31. Which metal salt gives a light violet colour in non-luminous and remain colourless in luminous flame under cold condition in borax bead test?

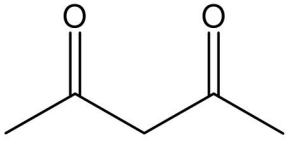
A) Fe^{+2} B) Ni^{+2} C) Co^{2+} D) Mn^{+2}

32. The compound that gives negative Fehling's test and +ve Iodoform test are:

(a) 

(b) 

(c) $CH_3-\overset{\overset{O}{\|}}{C}-H$

(d) 

(e) $Ph-\overset{\overset{O}{\|}}{C}-CH_3$

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Choose the correct answer from the options given below:

A) b & d only B) b, c, d and e only

C) b, d and e only D) b, c and d only

33. Which of the following set of quantum numbers is not allowed for an electron in an atom

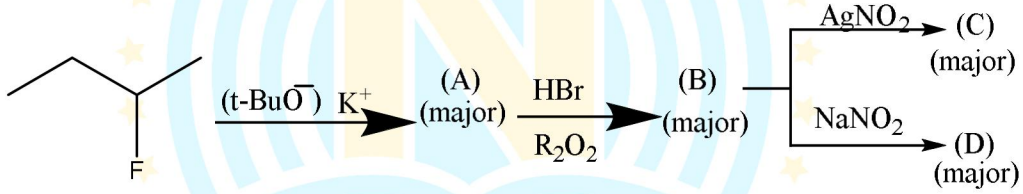
(a) $n=3,\ell=2,m=0$ (b) $n=2,\ell=2,m=-1$

(c) $n=3,\ell=0,m=1$ (d) $n=5,\ell=2,m=-1$

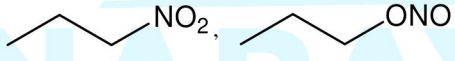
Choose the correct answer from options given below

A) a, b only B) b, c only C) c, d only D) a, b, c only

34.



The products D & C in the following reactions respectively

A) 

B) 

C) 

D) 

35. For a given aqueous solution of urea, the difference between boiling point and melting point is 104.76 K. The weight in gram of urea present in 100gm of water is_____ (for water $K_f=1.86, K_b=0.52$)

A) 60 gm B) 12 kg C) 12 gm D) 120 gm

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36. Choose the correct combination (s) for different Xe compounds about the hybridisation of Xe and lone pair of e^- on Xe in the compounds.

a) $XeO_2F_2 - sp^3d$ – one lone pair

b) $XeOF_4 - sp^3d^2$ – one lone pair

c) $XeO_3 - sp^3$ – one lone pair

d) $XeF_6 - sp^3d^3$ – one lone pair

A) a and d only

B) c and d only

C) a, b, c and d

D) a, b and c only

37. The reaction between X and Y is first order with respect to X and zero order with respect to Y. The following experimental data is observed for the reaction

Expt	$[X]$ mol L^{-1}	$[Y]$ mol L^{-1}	Rate mol L^{-1} min^{-1}
I	0.1	0.1	2×10^{-3}
II	L	0.2	4×10^{-3}
III	0.4	0.4	$M\times10^{-3}$
IV	0.1	0.2	2×10^{-3}

Consider the following statements for the above reaction

(a) The rate constant of the reaction is $2\times10^{-2}\ min^{-1}$

(b) The ratio of M to L is 40

(c) In a time of 34.65 minutes $[X]$ reacted is 50% of its initial concentration

Choose the correct answer from the flowing options

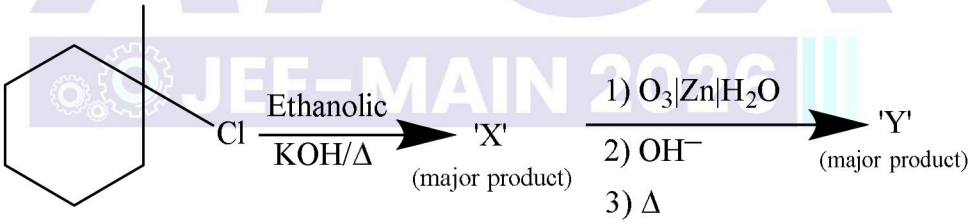
A) a, b and c

B) b and c only

C) a and c only

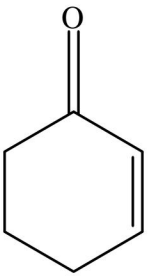
D) a and b only

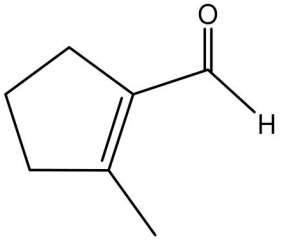
38.

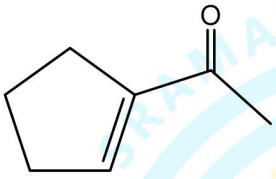


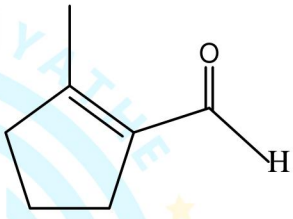
Major product 'Y' is

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A) 

B) 

C) 

D) 

39. Which of the following statement is wrong for the given equilibria

A) Solubility of a gas in water decreases with increase in temperature

B) $CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$, equilibrium shifts in forward direction if the reaction is carried out in an open vessel

C) $H_2O(g) \rightleftharpoons H_2O(\ell)$, the equilibrium will shift to right upon increasing temperature

D) $CO_2(g) \rightleftharpoons CO_2(\ell)$, the equilibrium will shift to right upon increasing pressure

40. Among the following molecules: SO_2, SF_4, ClF_3, BrF_5 and XeF_4. Which of the following shapes does not describe any of the molecules mentioned?

A) Bent B) Trigonal bipyramidal

C) See-Saw D) T-Shape

41. The ionic radius (A°) of C^{4-} and O^{2-} respectively are 2.60 and 1.40. The ionic radius of the isoelectronic ion N^{3-} would be

A) 2.60 B) 1.71 C) 1.40 D) 0.95

42. Given below are two statements

Statement I: In permanganate titrations, permanganate ion MnO_4^- acts as the self indicator

Statement II: In Dichrometry, $Cr_2O_7^{-2}$ is not a self indicator

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Choose the correct answer from the options given below

A) Statement I is false but Statement II is true

B) Both Statement I and Statement II are true

C) Statement I is true but Statement II is false

D) Both Statement I and Statement II are false

43. Match the List-I with List-II

	List-I (Reaction)		**List-II (Types)**
A)	$P_4 + 3OH^- + 3H_2O \rightarrow PH_3 + H_2PO_2^-$	I)	Redox reaction and Comproportionation
B)	$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$	II)	Not a Redox reaction
C)	$NH_4NO_3 \rightarrow N_2O + 2H_2O$	III)	Redox reaction and metal displacement
D)	$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$	IV)	Redox reaction and disproportionation

Choose the correct answer from the options given below

A) A-IV, B-III, C-II, D-I

B) A-III, B-II, C-IV, D-I

C) A-IV, B-I, C-II, D-III

D) A-IV, B-III, C-I, D-II

44. Statement-I: Allylic chloride exhibits more reactivity for S_N^1 reaction than vinyl chloride.

Statement-II: Unimolecular nucleophilic substitution reaction involves a carbocation intermediate.

In the light of the above statements, choose the most appropriate answer from the options given below

A) Both statement I and II are correct

B) Statement I is incorrect but statement II is correct

C) Both statement I and II are incorrect

D) Statement I is correct but statement II is incorrect

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45. Choose the correct options among the following

1) pH of an aqueous solution with $\left[H^+\right]=10^{-8}M$ is equal to 8

2) pH of 0.1M weak acid HA is equal to 4 and when it is diluted by 100 times its pH is increased to 5

3) pH of a mixture of $1.0\,ml$, $10^{-3}M\ HCl$ and $999\,ml$ of $10^{-3}M\ NaCl(aq)$ solution is equal to 6

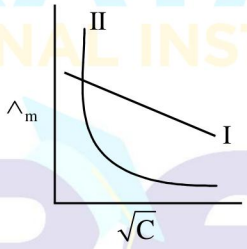
4) pOH of $10^{-8}M\ NaOH$ solution is equal to 7.04

A) 2 and 3 only
B) 1, 2, 3 and 4
C) 1, 2 and 3 only
D) 2, 3, 4 only

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

46. Given below is the plot of molar conductivity vs $\sqrt{C}$



Among the following the number of species having similar graph of I is x and II is y. Then the value of $(x-y)$ is____

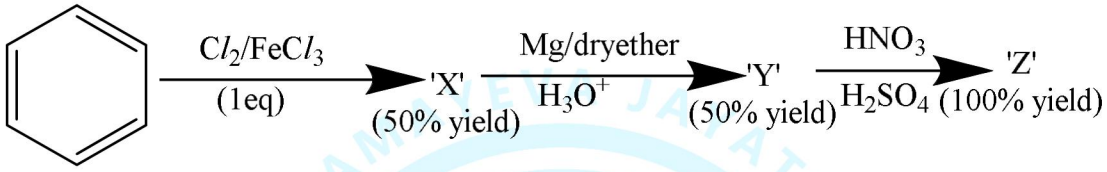
$AgNO_3, CH_3COOH, KCl, HCl, NH_4OH, H_3PO_4, MgSO_4,$
C_6H_5COOH, CH_3COONa

47. In a compound, C, H and N atoms are present in 9 : 1 : 3.5 by weight. Molecular weight of compound is 108. Molecular formula of a compound is $C_xH_yN_z$. Then the value of $(y+z+x)$ is___

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48. Density of a solution is 2 g/cc, which contains equal weights of solute and solvent. The ratio of molality to that of molarity is (x/y). Then the value of (x + y) is_____

49.



The Gram Molecular Weight (GMW) of the compound 'Z' is_____

50. The bond energies of $N \equiv N$ and $H-H$ are 945 and 435 kJ/mole respectively. If enthalpy of formation of NH_3 is $-50\,kJ/mol$, then the value of $N-H$ bond energy in ammonia in kJ/mol is___ (nearest integer)

MATHS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. The number of different 6 digit numbers greater than 500000 that can be formed using the digits 0, 1, 2, 3, 4, 5, 6, 7, such that the sum of their first and last digits should not be more than 8, is N then number of positive integral divisors of (N+1) is

A) 36 B) 39 C) 52 D) 56

52. Let ABCD be a trapezium whose vertices lie on the parabola $y^2 = 8x$. Let the sides AD and BC of the trapezium be parallel to y-axis. If the diagonal AC is of length $\frac{25}{2}$ and it passes through the point (2, 0), then the area of ABCD is :

A) 75 B) 25 C) 125 D) $\frac{75}{2}$

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53. Let $f(x)=\cos x$. *If* a real value of 'x' is chosen at random on the interval $-12\pi < x < 10\pi$, then the chance that $f(x)$ is positive is

A) $\frac{1}{3}$ B) $\frac{1}{2}$ C) $\frac{1}{4}$ D) $\frac{1}{\sqrt{2}}$

54. Let $f(k)=\frac{k}{2009}$ *and* $g(k)=\frac{f^4(k)}{(1-f(k))^4+(f(k))^4}$ then $\sum_{k=0}^{2009} g(k)$ is equal to

A) 1010 B) 2020 C) 1005 D) 2005

55. Let $f(x)$ be a function such that $f:Q\to Q$ ('Q' is set of rational numbers)

$f(xy)=f(x).f(y)-f(x+y)+1 \quad \forall x,y\in Q$

$f(1)=2$ then $1+\sum_{K=1}^{\infty}\frac{f(K)}{2^K}$ is

A) 2 B) 4 C) 8 D) 16

56. Let $A(x,y,z)$ be a point in XY plane which is equidistant from $(1,-1,0)$, $(2,1,2)$ and $(3,2,-1)$. Let $B=(1,3,-1)$ and $C=(2,0,2)$, then consider the following statements:

S–1: ΔABC is an isosceles triangle

S–2: Centroid of ΔABC is $\left(\frac{3}{2},\frac{4}{3},\frac{1}{3}\right)$

A) Both statements are true B) only S – 1 is true

C) only S – 2 is true D) Both statements are false

57) Let a relation R in the set N of natural numbers be defined as $(x,y)\Leftrightarrow x^2-4xy+3y^2=0$ *for all* $x,y\in N$ The relation R is

A) Reflexive but not symmetric and not transitive

B) Symmetric but not reflexive and not transitive

C) Transitive but not reflexive and not symmetric

D) Equivalence relation

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58. Centre of the circle touching y-axis $at(0,3)$ and making an intercept '2' units on positive $x-axis$ is

A) $\left(\sqrt{10},3\right)$ B) $\left(3,\sqrt{10}\right)$ C) $\left(-\sqrt{10},3\right)$ D) $\left(-\sqrt{10},-3\right)$

59. The sum to infinity of the series

$$1+\left(1+\frac{1}{2}\right)\left(\frac{1}{3}\right)+\left(1+\frac{1}{2}+\frac{1}{2^2}\right)\left(\frac{1}{3^2}\right)+.....\propto is$$

A) $\frac{12}{5}$ B) $\frac{9}{5}$ C) $\frac{9}{10}$ D)3

60. If the mapping $f(x)=ax+b, a>0$ maps [-1,1] onto [0,2] then $\cot\left[\cot^{-1}7+\cot^{-1}8+\cot^{-1}18\right]$ is___

A) $f(-1)$ B) $f(0)$ C) $f(1)$ D) $f(2)$

61. If T_r be the rth term of a sequence, for $r=1,2,3-----$. *If* $3T_{r+1}=T_r$ and $T_7=\frac{1}{243}$, then the value of $\sum_{r=1}^{\infty}T_r.T_{r+1}$ is

A) $\frac{29}{8}$ B) $\frac{27}{8}$ C) $\frac{25}{8}$ D) $\frac{21}{8}$

62. Let $P(\alpha,\beta,\gamma)$ be the image of the $Q(1,2,3)$ in the line $\frac{x-0}{1}=\frac{y-3}{1}=\frac{z-1}{-1}$ and R be the point $(2,5,-1)$. Then the area of the triangle PQR is

A) $\frac{2\sqrt{14}}{3}$ B) $\frac{4\sqrt{14}}{3}$ C) $2\sqrt{14}$ D) $\frac{8\sqrt{14}}{3}$

63. Consider a function $f(x)$ which satisfies $f'(x)=tan^2x+\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}}f(x)dx$. Also $f\left(\frac{\pi}{4}\right)=-\frac{\pi}{4}$ then $\frac{\left(8+\pi^2\right)}{\pi}\int_{\frac{-\pi}{4}}^{\frac{\pi}{4}}f(x)dx$ is

A) – 2 B) – 3 C) – 4 D) – 6

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64. Consider $f(x)=\begin{cases} x^3-3x+3 & :-2\le x<0 \\ x+1 & :0\le x\le 5 \\ 1-e^x & :x>5 \end{cases}$

A) Sum of two local maximum values is 9

B) $f(x)$ has no local minima C) Global minima of $f(x)$ is $1-e^5$

D) Sum of two local maximum values is 11

65. The set of all real numbers $'x'$ for which $x^2-|x+2|+x>0$ is

A) $(-\infty,-2)\cup(2,\infty)$ B) $(-\infty,-\sqrt{2})\cup(\sqrt{2},\infty)$

C) $(-\infty,-1)\cup(1,\infty)$ D) $(\sqrt{2},\infty)$

66. Let $F:[3,5]\to R$ be a twice differentiable function on (3,5) such that

$F(x)=e^{-x}\int_3^x\left(3t^2+2t+4F'(t)\right)dt$ Then $F(x)=$ _____

A) $\frac{x^3-x^2-36}{e^x+4}$ B) $\frac{x^3+x^2-36}{e^x+4}$ C) $\frac{x^3+x^2-36}{e^x-4}$ D) $\frac{x^3+x^2+36}{e^x-4}$

67. Let ${}^nC_{r-1}=4$, ${}^nC_r=12$ and ${}^nC_{r+1}=16$. Let $A(\cos t,\sin t)$, $B(\sin t,-\cos t)$ and $C(5(n-r),5(n+r))$ be the vertices of a triangle ABC where, t is a parameter. If $(3x-16)^2+(3y-30)^2=\alpha$ is the locus of the centroid of triangle ABC, then α equals to____

A) 14 B) 3 C) 2 D) 1

68. Let Z_1,Z_2 & Z_3 be complex numbers satisfying $|Z-i\sqrt{3}|=1$ and $3Z_1+i\sqrt{3}=2Z_2+2Z_3$ then $|Z_1-Z_2|$ is

A) $\frac{1}{\sqrt{2}}$ B) 2 C) $\sqrt{2}$ D) $2\sqrt{2}$

69. Two cards are drawn simultaneously (or successively without replacement) from a well shuffled pack of 52 cards. The variance of the number of kings is σ^2 then $(221)^2\sigma^2$ is

A) 3500 B) 4800 C) 5500 D) 6800

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70) Let $S=\left\{(x,y)\in R\times R: x\geq 0, y\geq 0, y^2\leq 4x, y^2\leq 12-2x \text{ and } 3y+\sqrt{8}x\leq 5\sqrt{8}\right\}$.

If the area of the region S is $a\sqrt{2}$, then α is equal to

A) $\frac{17}{2}$ B) $\frac{17}{3}$ C) $\frac{17}{4}$ D) $\frac{17}{5}$

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
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71. Let M denote the set of all real metrices of order 3×3 and let

$S=\{-3,-2,-1,0,1,2,3\}$

Let $S_1=\left\{A=\left[a_{ij}\right]\in M: A=A^T \text{ and } a_{ij}\in S,\forall i,j\right\}$

$S_2=\left\{A=\left[a_{ij}\right]\in M: A=-A^T \text{ and } a_{ij}\in S,\forall i,j\right\}$

$S_3=\left\{A=\left[a_{ij}\right]\in M, a_{11}+a_{22}+a_{33}=10 \text{ and } a_{ij}\in S,\forall i,j\right\}$

Then the remainder when $n\left(S_1\cup S_2\cup S_3\right)$ is divided by 7 is

72. If $S={}^{404}C_4-{}^4C_1.{}^{303}C_4+{}^4C_2.{}^{202}C_4-{}^4C_3.{}^{101}C_4=(101)^k$ then the perpendicular distance from (0,K) to the straight line $\sqrt{3}x-y-2=0$ is

73. Let $\bar{u},\bar{v}$ are two non zero vectors, $\bar{u}=\bar{v}+2\left(\bar{v}\times\hat{i}\right)$ where $\bar{v}=v_1\hat{i}+v_2\hat{j}+v_3\hat{k}$ and v_1,v_2,v_3 are non negative integers which are distinct from each other then minimum value of $|\bar{u}|$ is

74. Let $f(x)=\begin{cases}1-|x|, & |x|\leq 1\\ 0, & |x|>1\end{cases}$ and $g(x)=f(x-1)+f(x+1)$. Then the number of points where g is not differentiable is

75. An ellipse has OB as semi minor axis, F and F' are its focii and the angle FBF' is a right angle, and the eccentricity of ellipse is $\frac{\alpha}{\sqrt{\beta}}$ (where α,β are co-prime). Then the value of $\left|\alpha+\beta^2\right|$ is equal to____

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